

standard deviation of the S&P 500 excess return. I have also de-measured all series. The panels in Figure 17.6 represent very different HF styles: the HF index in Panel A, long/short equity in Panel B, merger arbitrage in Panel C, and relative value in Panel D. Remarkably, they all show a general pattern of steady and relatively high returns during the early 2000s, a substantial drawdown during 2008, and a recovery after 2009. There is an amazingly close correspondence of these HF returns with the S&P 500 and volatility factors in all the panels. *HFs repackage equity and volatility risk.*

Table 17.5 and Figure 17.6 take HF returns aggregated to the whole industry or the sector level. At the individual HF level, the literature also finds that volatility risk factors explain a very large part of HF returns. Often any outperformance of HFs, if it exists, can be partly and sometimes wholly attributed to volatility risk exposure. Fung and Hsieh (2001), the first to recognize the importance of volatility risk factors in HF returns, find no outperformance of HFs once volatility risk factors are included.³³ Fung et al. (2008) find an average HF alpha of just 6 basis points (effectively zero) with a risk benchmark that includes equity, bond and volatility risk factors.

Other Factors

There are other important factors that drive HF returns besides the equity market and volatility factors, but they tend to be sector-specific. Commodities and currency risk, for example, are reflected in macro HF returns. (Commodities and currencies themselves also reflect volatility risk.) Term spreads (the difference between long and short maturity Treasury bonds) and credit spreads (the difference between risky and safe corporate debt) are also significant risk factors for relative value strategies. Quant funds are highly exposed to value-growth and momentum risk (covered in chapter 7), and not surprisingly these factors didn't do well during August 7 to 9, 2007.

Sadka (2010) finds that many HFs are exposed to illiquidity risk. HFs with the highest illiquidity risk exposures outperform their more liquid counterparts by 6%. Interestingly, this illiquidity risk premium is not related to how easily investors can withdraw money from the HFs themselves (i.e., the illiquidity characteristics of the HFs; see below). But the most important risk factors are regular market risk and volatility risk.

In 2004, Asness foretold:

In today's world, these strategies are not only linked by a common risk-taking/liquidity-providing element, but are also more and more pursued by the exact same investors. It is easy to imagine considerably greater potential co-movement in a crisis because of this commonality.

³³ See also Agarwal and Naik (2004), Fung and Hsieh (2004), and Lo and Hasanhodzic (2007).